Chapter 15: Gnathiferans and Smaller Lophotrochozoans

Introduction to Lophotrochozoans

- **Protostomia** is a large clade divided into two groups: **Lophotrochozoa** and **Ecdysozoa**.
- Lophotrochozoan embryos typically exhibit **spiral mosaic cleavage**.
- Body plans within Lophotrochozoa vary:
  - **Acoelomate**: No body cavity (e.g., Platyhelminthes, Gnathostomulida).
  - **Pseudocoelomate**: Internal cavity surrounding the gut (**pseudocoel**) not completely lined with mesoderm; derived from the embryonic **blastocoel** (also called **blastocoelomates**).
  - **Eucoelomate** (Coelomate): True coelom completely lined with mesoderm.
- Functions of the **pseudocoel**:
  - Space for organ development.
  - Simple circulation/distribution of materials.
  - Storage of waste products.
  - **Hydrostatic support** (hydrostatic skeleton).
- Many pseudocoelomates are small and rely on the pseudocoel for internal circulation in the absence of a true circulatory system.

Figure 1: Acoelomate, pseudocoelomate, and eucoelomate body plans. Acoelomates lack a body cavity; pseudocoelomates have a cavity derived from the blastocoel, partially lined by mesoderm; eucoelomates have a true coelom completely lined by mesoderm.

- **Figure 15.1 Explanation**: Comparison of body plans. Acoelomates have a solid body filled with parenchyma. Pseudocoelomates have a cavity (pseudocoel) between the endoderm (gut) and mesoderm (muscle). Eucoelomates have a cavity (coelom) fully enclosed by mesoderm (peritoneum).

Clade Gnathifera

- Includes four phyla: **Gnathostomulida**, **Micrognathozoa**, **Rotifera**, and **Acanthocephala**.
- Characterized by small **cuticular jaws** with a homologous microstructure (except Acanthocephala, which have lost them).
- **Syndermata**: A proposed clade uniting **Rotifera** and **Acanthocephala** based on molecular data and the shared presence of a **eutelic syncytial epidermis**.
  - **Eutely**: Constancy in the number of nuclei/cells in the body or organs.

Figure 2: Hypothetical relationships among gnathiferans and some lophotrochozoans. The cladogram shows relationships based on morphological and molecular characters, highlighting the position of Gnathifera (Gnathostomulida, Micrognathozoa, Rotifera, Acanthocephala) and other lophotrochozoans.

- **Figure 15.2 Explanation**: Phylogenetic tree showing Gnathifera as a clade within Lophotrochozoa. It highlights the relationship between Rotifera and Acanthocephala (Syndermata) and the position of other smaller lophotrochozoans like Gastrotricha and the lophophorates.

Phylum Gnathostomulida (Jaw Worms)

- Tiny, wormlike animals (< 2 mm) living in interstitial spaces of sand/silt (**meiofauna**).
- Can endure low oxygen conditions.
- **Epidermis** is ciliated, but each cell has only **one cilium** (monociliated).
- Feed by scraping bacteria/fungi with a pair of **jaws** on the pharynx.
- Digestive system: Simple blind gut (no anus).
- Body plan: **Acoelomate**. No circulatory or respiratory systems.
- Reproduction: Mostly **hermaphroditic** (protandric or simultaneous).

Figure 3: A, **Gnathostomula jenneri** (phylum Gnathostomulida), a tiny interstitial jaw worm found in shallow water and sediments. B, **Gnathostomula paradoxa**, abundant in sediments near marine polychaete burrows.

- **Figure 15.3 Explanation**: **Gnathostomula** species are typical jaw worms found in marine sediments. They are delicate and worm-like.

Phylum Micrognathozoa

- Represented by a single described species: **Limnognathia maerski**.
- Tiny, interstitial animals found in Greenland.
- Body: Two-part head, thorax, abdomen, short tail.
- Locomotion: Cilia and a unique **ventral ciliary adhesive pad**.
- Feeding: Three pairs of complex **jaws**.
- Gut: Simple, with an anus that opens periodically.

Figure 4: A, **Limnognathia maerski**, the only known species of Micrognathozoa. B, Detail of its complex jaws. C, A living specimen found on moss in a freshwater spring in Greenland.

- **Figure 15.4 Explanation**: **Limnognathia maerski** has a complex jaw apparatus, visible in the detail view, which is a key characteristic of Gnathifera.

Phylum Rotifera

- Name means “wheel-bearer”, referring to the ciliated **corona**.
- Mostly freshwater, some marine, terrestrial, or parasitic.

- Can endure desiccation (**cryptobiosis**) for long periods.

Form and Function

- Body regions: Head (with corona), Trunk, Posterior tail/foot.
- **Corona**: Ciliated crown used for locomotion and feeding.
- **Mastax**: Muscular pharynx equipped with hard jaws called **trophi**.
- **Lorica**: Thickened, case-like fibrous layer in the epidermis of some species.
- **Foot**: Attachment organ, often with **toes** and **pedal glands** (secrete adhesive).
- **Syncytial epidermis**: Secretes the cuticle; nuclei are constant in number (**eutely**).
- **Pseudocoel**: Large, fluid-filled, provides hydrostatic support.
- Digestive system: Complete. Stomach absorbs nutrients.
- Excretory system: Pair of **protonephridia** with **flame cells**; empty into a **cloaca** (bladder).
- Nervous system: Bilobed **brain**, eyespots, sensory bristles.

Figure 5: A, A live **Philodina**, a common rotifer. B, Structure of **Philodina**, showing the corona, mastax, digestive system, and foot with spurs and toes.

- **Figure 15.5 Explanation**: **Philodina** illustrates the typical rotifer anatomy: ciliated corona for feeding/swimming, mastax for grinding food, and a foot with toes for attachment.

Figure 6: Variety of form in rotifers. A, **Stephanoceros** with fingerlike coronal lobes. B, **Asplanchna**, a pelagic predator. C, **Squatinella** with a hoodlike extension. D, **Macrochaetus**, dorsoventrally flattened.

- **Figure 15.6 Explanation**: Rotifers exhibit great morphological diversity. **Stephanoceros** is sessile with trapping lobes; **Asplanchna** is a sac-like pelagic predator; **Macrochaetus** is flattened.

Reproduction

- Dioecious, but males are often smaller or unknown (e.g., Class Bdelloidea).
- **Parthenogenesis** is common.
- Class Monogononta reproductive cycle:
  - **Amictic eggs**: Diploid, thin-shelled, develop parthenogenetically into diploid females. Produced during favorable conditions.
  - **Mictic eggs**: Haploid, produced by **mictic females** in response to environmental stress.
    - If unfertilized: Develop into haploid **males**.
    - If fertilized: Develop into **dormant eggs** (thick-shelled, diploid) that survive winter and hatch into amictic females.

Figure 7: Reproduction of some rotifers (class Monogononta). The cycle includes amictic (diploid) parthenogenesis during favorable conditions and mictic (sexual) reproduction triggered by environmental stimuli, producing dormant eggs.

- **Figure 15.7 Explanation**: The rotifer life cycle alternates between asexual (amictic) reproduction and sexual (mictic) reproduction. Dormant eggs are the product of sexual reproduction and allow survival through harsh conditions.

Phylum Acanthocephala (Spiny-Headed Worms)

- All are **endoparasites** in the intestine of vertebrates (fish, birds, mammals).
- Larvae develop in arthropods (crustaceans or insects).
- Key feature: Cylindrical, invaginable **proboscis** bearing rows of recurved **spines** for attachment.

Form and Function

- Body wall: **Syncytial tegument** with a **lacunar system** of fluid-filled canals (facilitates diffusion).
- No digestive tract; nutrients absorbed across the tegument.
- **Lemnisci**: Hydraulic sacs in the neck region, likely reservoirs for lacunar fluid.
- Excretion: Protonephridia (if present).
- Reproduction: Dioecious.
  - Females have **ovarian balls** floating in the pseudocoel.
  - **Uterine bell**: Sorting apparatus that allows only mature, shelled embryos to pass into the uterus and out with host feces; immature eggs are returned to the body cavity.

Figure 8: Structure of a spiny-headed worm (phylum Acanthocephala). A and B, Eversible spiny proboscis for attachment. C, Live **Leptorhynchoides thecatus**. D, Male (smaller). E, Female genital selective apparatus separating immature from mature eggs.

- **Figure 15.8 Explanation**: The spiny proboscis (A, B) anchors the worm in the host gut. The female reproductive system (E) features a unique uterine bell that sorts eggs based on maturity.

Phylum Cycliophora

- Discovered in 1995 on mouthparts of lobsters (**Symbion pandora**).
- Acoelomate body; cellular epidermis.
- Feeding: Ring of compound cilia around the mouth; U-shaped gut.
- Complex life cycle:
  - **Pandora larva**: Asexual, forms new feeding stage.
  - **Chordoid larva**: Sexual dispersal stage, swims to new hosts.

Figure 9: **Symbion pandora**, a cycliophoran living on setae on the mouthparts of lobsters. It has a complex life cycle with sexual and asexual phases.

- **Figure 15.9 Explanation**: **Symbion pandora** is a tiny commensal/symbiont on lobster setae, with a complex life cycle involving internal budding.

Phylum Gastrotricha

- Small, ventrally flattened animals; glide on ventral cilia.
- Dorsal surface often has bristles, spines, or scales.
- **Acoelomate**.
- **Adhesive tubes**: Dual-gland system for attachment.
- Digestive system: Complete.
- Excretion: Protonephridia with **solenocytes** (single flagellum) instead of flame cells.
- Reproduction: Hermaphroditic or parthenogenetic.

Figure 10: A, Live **Chaetonotus simrothic**, a common gastrotrich. B, Dorsal surface showing scales/spines. C, Internal structure, ventral view, showing the pharynx, intestine, and adhesive tubes.

- **Figure 15.10 Explanation**: **Chaetonotus** shows the typical bristly appearance. Internally, it has a pharynx, straight intestine, and adhesive tubes.

Figure 11: Gastrotrichs in order Macrodasyida. A, **Macrodasys**. B, **Turbanella**. These are elongated forms often found in marine interstitial habitats.

- **Figure 15.11 Explanation**: Macrodasyid gastrotrichs like **Macrodasys** are more elongated and worm-like compared to the chaetonotids.

Phylum Entoprocta

- Tiny, sessile, stalked animals; solitary or colonial.
- Body consists of a cup-shaped **calyx** and a crown of ciliated tentacles.
- Key distinction: Both **mouth and anus** open **within** the circle of tentacles.
- Feeding: **Ciliary suspension feeders**.
- **Pseudocoelomate** (filled with gelatinous parenchyma).
- Larva: Trochophore-like.

Figure 12: A, **Urnatella**, a freshwater entoproct forming small colonies. B, **Loxosomella**, a solitary entoproct. C, A live **Loxosomella**. Note the cup-shaped calyx and tentacular crown.

- **Figure 15.12 Explanation**: Entoprocts resemble hydroids or bryozoans but are distinct. **Urnatella** is a freshwater genus. The tentacles roll inward for protection.

Lophophorates

- Includes phyla **Ectoprocta**, **Brachiopoda**, and **Phoronida**.
- Possess a **lophophore**: Crown of ciliated tentacles borne on a ridge/fold, used for feeding and respiration.
  - Mouth is **inside** the ring; anus is **outside**.
- Coelom is tripartite (divided into three parts):
  - **Protocoel** (epistome).
  - **Mesocoel** (lophophore).
  - **Metacoel** (trunk/viscera).
- Phylogenetic position: Lophotrochozoa (protostomes), though they share some deuterostome-like features (radial cleavage, enterocoely in some).

Figure 13: A, Skeletal remains of a colony of **Membranipora**, a marine encrusting bryozoan (Ectoprocta). B, Portion of a colony showing extended zooids. The mouth is inside the lophophore, anus outside.

- **Figure 15.13 Explanation**: Ectoprocts (Bryozoans) live in secreted chambers (**zoecia**). The animal (**zooid**) consists of the **polypide** (feeding part) and **cystid** (body wall/case).

Phylum Ectoprocta (Bryozoa)

- "Moss animals". Mostly marine, colonial, encrusting or erect.
- **Zooid**: The individual colony member.
- **Zoecium**: The exoskeleton (gelatinous, chitinous, or calcareous).
- **Polypide**: Lophophore, digestive tract, muscles, nerves.
- **Cystid**: Body wall and exoskeleton.
- Feeding: Lophophore extends by hydrostatic pressure; retracts by muscles.
- Polymorphism:
  - **Avicularium**: Beak-like zooid for defense.
  - **Vibraculum**: Bristle-like zooid for sweeping.
- Reproduction:
  - Hermaphroditic.
  - Freshwater species produce **statoblasts**: Resistant, dormant capsules for overwintering.

Figure 14: Colonies of marine ectoprocts. A, Lacy colony of **Membranipora tuberculata**. B, **Bugula neritina** with upright, branching colonies.

- **Figure 15.14 Explanation**: Ectoproct colonies vary in form, from encrusting sheets (**Membranipora**) to upright branching structures (**Bugula**).

Figure 15: A, Ciliated lophophore of **Electra pilosa**, a marine ectoproct. Also shows a vibraculum. B, **Plumatella repens**, a freshwater bryozoan with a U-shaped lophophore.

- **Figure 15.15 Explanation**: The lophophore is circular in marine species (A) and often U-shaped in freshwater species (B).

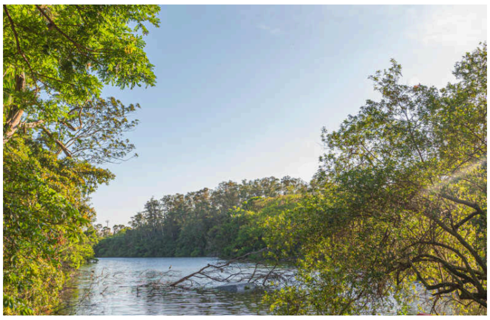

Figure 16: Statoblast of a freshwater ectoproct **Cristatella**. This resistant capsule with hooked spines allows survival during harsh conditions.

- **Figure 15.16 Explanation**: Statoblasts are unique to freshwater ectoprocts, functioning like seeds to regenerate the colony after winter.

Phylum Brachiopoda (Lamp Shells)

- Marine, bottom-dwelling.
- Resemble bivalves but have **dorsal and ventral valves** (shells), not lateral.
- Attached to substrate by a fleshy stalk called a **pedicel**.
- Two classes:
  - **Articulata**: Valves connected by a tooth-and-socket hinge.
  - **Inarticulata**: Valves held by muscles only (e.g., **Lingula**).
- Feeding: Large lophophore inside the shell creates currents.

Figure 17: Brachiopods. A, **Lingula**, an inarticulate brachiopod living in a burrow. B, **Terebratella**, an articulate brachiopod with a tooth-and-socket hinge and a short pedicel.

- **Figure 15.17 Explanation**: **Lingula** (A) is a living fossil (inarticulate). **Terebratella** (B) is an articulate brachiopod with a typical lamp-shell shape.

Figure 18: Phylum Brachiopoda. A, Longitudinal section of an articulate brachiopod showing dorsal/ventral valves and lophophore. B, Feeding and respiratory currents over the lophophore.

- **Figure 15.18 Explanation**: Internal anatomy shows the lophophore occupying much of the mantle cavity. The pedicel extends from the ventral valve.

Phylum Phoronida

- Wormlike, live in leathery/chitinous **tubes** in shallow water.
- **Horseshoe-shaped lophophore** with spiraled horns.
- Closed circulatory system with hemoglobin.
- Larva: **Actinotroch**.

Figure 19: Internal structure of **Phoronis** (phylum Phoronida), diagrammatic vertical section showing the U-shaped gut, lophophore, and tube.

- **Figure 15.19 Explanation**: Phoronids have a U-shaped gut with the anus dorsal to the mouth, outside the lophophore. They live within a secreted tube.

Phylogeny Summary

- **Lophotrochozoa** includes acoelomates, pseudocoelomates, and coelomates.
- **Gnathifera** (Gnathostomulida, Micrognathozoa, Rotifera, Acanthocephala) share jaw structure (or ancestry).
- **Acanthocephala** are likely highly derived **Rotifera** (clade **Syndermata**).
- **Lophophorates** (Ectoprocta, Brachiopoda, Phoronida) share the lophophore and tripartite coelom but their monophyly is debated; molecular data places them in Lophotrochozoa.